

Rashtreeya Sikshana Samithi Trust
RV UNIVERSITY
School of Computer Science and Engineering
Bengaluru – 560059



**Incident Response and Digital
Forensics**

COURSE CODE: CS2406

IV SEMESTER B.Tech (Hons.)

LABORATORY MANUAL

2025-2026

RV UNIVERSITY
School of Computer Science and Engineering
Bengaluru – 560 059



LABORATORY CERTIFICATE

This is to certify that Mr./Ms. _____
has satisfactorily completed the course of experiments in Practical *Incident Response and Digital Forensics* (CS2406) prescribed by the School of Computer Science and Engineering during the year 2025-26.

Name of the Candidate: _____

USN: _____ Semester: _____

Marks	
Maximum	Obtained
10	

Signature of Faculty in-charge

Program Director

Date:

Vision and Mission of the School of Computer Science and Engineering

Vision

To be a pioneering school of Computer Science and Engineering committed to fostering liberal education and empowering the next generation of technologists to make a positive global socio-economic impact.

Mission

- To be a pioneer in computer science education and benchmark ourselves with the world's top computer science and engineering institutions.
- To provide state-of-the-art facilities that enable exemplary pedagogy, advanced research, innovation and entrepreneurship in emerging technologies of computer science.
- To promote a culture of cooperation and inclusiveness among students and faculty from diverse communities enabling them to take part in interdisciplinary and multidisciplinary research, contributing to institution-building.
- To foster excellence through national and international academic, industry collaborations, bringing in diverse perspectives to drive innovation.
- To nurture a talented pool of ethical, self-driven and empathetic problem solvers to achieve sustainable development goals.

LIST OF LABORATORY PROGRAMS

Sl. No.	Title of the Experiment
1	Incident Response Simulation Using TheHive
2	Draft an Incident Response Plan Document
3	Develop an Incident Response Playbook for DDoS Attacks
4	Tabletop Exercise for Social Engineering Attack Response
5	Incident Ticket Management Using Jira or Request Tracker (RT)
6	Write a Forensic Report Based on Simulated Attack Artifacts
7	Reconstruct an Attack Timeline Using ELK Stack
8	Collect and Analyze Threat Intelligence from OTX and VirusTotal
9	Classify Threats Using MITRE ATT&CK Navigator
10	Prepare a Threat Intelligence Report for SOC Operations
11	Configure Open Source SIEM (Security Onion) to Use Threat Feeds
12	Threat Hunting with Zeek (Bro) IDS Using IOC Indicators

Lab Write-up and Execution rubrics (Max: 6 marks)					
Sl. No	Criteria	Measuring Methods	Excellent	Good	Poor
1	Understanding of problem statements (2 Marks)	Observations	Student exhibits a thorough understanding of the problem statements and applies appropriate concepts to devise a solution. (2 M)	Student has sufficient understanding of the problem statements and applies suitable concepts to devise a solution. (<2 M and >=1 M)	Student does not have a clear understanding of the problem statements and is unable to apply appropriate concepts to devise a solution. (0 M)
2	Execution (2 Marks)	Observations	Student demonstrates the execution of the program with optimized code, effectively utilizing principles and shows performance efficiency. All test cases are handled with appropriate validations. (2 M)	Student demonstrates the execution of the program with functional code, utilizing principles but without optimization. Performance efficiency is shown with only a few test cases handled. (1 M)	Student has not executed the program or the code fails to demonstrate correct use of principles. (0 M)
3	Results and Documentation (2 Marks)	Observations	Documentation with appropriate comments and output is covered in manual. (2 M)	Documentation with only few comments and only few output cases is covered in manual. (1 M)	Documentation with no comments and no output cases is covered in manual. (0 M)
Viva Voce Rubrics (Max: 4 marks)					
1	Conceptual Understanding (2 Marks)	Viva Voce	Explains thoroughly the core concepts and principles and fundamentals. (2 M)	Adequately explains the core concepts and principles and fundamentals with some understanding. (1 M)	Unable to explain the core concepts and basic principles and fundamentals. (0 M)

2	Application of Concepts (2 Marks)	Viva Voce	Insightful explanation and demonstration of appropriate techniques and practical applications for solving problems (1 M)	Sufficiently explains and demonstrates the use of techniques and practical applications for solving problems with some understanding. (0.5 M)	Unable to explain and demonstrate the use of techniques and practical applications for solving problems. (0 M)
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Experiment 1: Incident Response Simulation Using TheHive

Aim

To simulate a malware incident and manage the complete incident response lifecycle using TheHive platform.

Inputs

- Simulated malware alert
- Hostname, IP address, hash value
- Timestamp of detection

Tools Used

- TheHive (Open Source IR Platform)
- Cortex (optional)
- Virtual Machine (Ubuntu)

Dataset Required

- Sample malware indicators (hash, IP, domain)
- Predefined incident scenario document

Brief Theory

TheHive is an incident response platform that helps SOC teams track, investigate, and respond to security incidents using structured workflows and observables.

Procedure

1. Launch TheHive server.
2. Create a new case titled *“Malware Infection on HR System”*.

3. Add observables (IP, file hash, domain).
4. Assign severity, tags, and analyst.
5. Create tasks (containment, eradication, recovery).
6. Document findings and close the case.

Implementation (if any)

- Configuration of TheHive server
- Optional Cortex analyzer integration

Expected Results

- Successful creation and closure of an incident case
- Structured documentation of response activities

Conclusion

This experiment demonstrates structured incident handling using an industry-grade IR platform.

Experiment 2: Draft an Incident Response Plan Document

Aim

To prepare a formal Incident Response Plan (IRP) for a fictional organization.

Inputs

- Organization profile
- Regulatory requirements

Tools Used

- Microsoft Word / Google Docs

Dataset Required

- Sample incident categories
- Compliance guidelines (ISO 27001 / NIST)

Brief Theory

An Incident Response Plan defines procedures, roles, and responsibilities to manage security incidents effectively.

Procedure

1. Define scope and objectives.
2. Identify incident categories.
3. Assign IR roles (IR Manager, Analyst, Legal).
4. Define communication and escalation flow.
5. Include incident templates.

Implementation

- Document preparation only

Expected Results

- Complete IR plan document

Conclusion

A documented IR plan improves organizational readiness and compliance.

Experiment 3: Incident Response Playbook for DDoS Attacks

Aim

To design a detailed DDoS response playbook.

Inputs

- DDoS attack scenario
- Network architecture

Tools Used

- Text editor
- Diagram tools

Dataset Required

- Sample DDoS indicators

Brief Theory

A playbook provides step-by-step actions to respond consistently to specific attack types.

Procedure

1. Define detection criteria.
2. Identify mitigation steps.
3. Assign roles.
4. Define recovery actions.
5. Document post-incident review steps.

Expected Results

- Structured DDoS response playbook

Conclusion

Playbooks reduce response time and analyst errors during attacks.

Experiment 4: Tabletop Exercise for Social Engineering Attack

Aim

To conduct a role-based tabletop exercise for social engineering attack response.

Inputs

- Phishing scenario

Tools Used

- Whiteboard / Document

Dataset Required

- Email phishing sample

Brief Theory

Tabletop exercises test decision-making and coordination without live systems.

Procedure

1. Assign roles (SOC, HR, Management).
2. Present attack scenario.
3. Discuss response actions.
4. Identify gaps and improvements.

Expected Results

- Improved response coordination

Conclusion

Tabletop exercises enhance organizational preparedness.

Experiment 5: Incident Ticket Management in Jira / RT

Aim

To manage incident tickets lifecycle using a ticketing system.

Inputs

- Simulated security events

Tools Used

- Jira / Request Tracker (RT)

Dataset Required

- Incident descriptions

Brief Theory

Ticketing systems ensure accountability and tracking of security incidents.

Procedure

1. Create incident tickets.
2. Assign priority and owner.
3. Update status.
4. Close tickets with resolution notes.

Expected Results

- Proper ticket lifecycle management

Conclusion

Ticket systems enhance visibility and auditability.

Experiment 6: Write a Forensic Report Based on Simulated Artifacts

Aim

To analyze logs and produce a formal forensic report.

Inputs

- Log files
- System artifacts

Tools Used

- Log viewer
- Text editor

Dataset Required

- Authentication and access logs

Brief Theory

Forensic reports document evidence in a legally acceptable manner.

Procedure

1. Analyze logs.
2. Identify suspicious activity.
3. Extract evidence.
4. Prepare report sections.

Expected Results

- Professional forensic report

Conclusion

Documentation is critical for legal and compliance needs.

Experiment 7: Attack Timeline Reconstruction Using ELK Stack

Aim

To visualize an attack timeline using ELK Stack.

Inputs

- Server logs

Tools Used

- Elasticsearch
- Logstash
- Kibana

Dataset Required

- Web and system logs

Brief Theory

ELK Stack enables centralized log analysis and visualization.

Procedure

1. Ingest logs using Logstash.
2. Index data in Elasticsearch.
3. Create timeline dashboards in Kibana.

Expected Results

- Visual attack timeline

Conclusion

ELK improves threat visibility and investigation.

Experiment 8: Threat Intelligence Collection from OTX & VirusTotal

Aim

To collect and analyze threat intelligence data.

Inputs

- Malicious IPs/domains

Tools Used

- AlienVault OTX
- VirusTotal

Dataset Required

- IOC list

Brief Theory

Threat intelligence provides context about known threats.

Procedure

1. Query IOCs in OTX.
2. Analyze VirusTotal reports.
3. Correlate findings.

Expected Results

- Threat context report

Conclusion: Threat intel improves proactive defense.

Experiment 9: MITRE ATT&CK Classification

Aim

To map threats to MITRE ATT&CK framework.

Inputs

- Indicators and behaviors

Tools Used

- MITRE ATT&CK Navigator

Dataset Required

- IOC list

Brief Theory

MITRE ATT&CK standardizes adversary behavior classification.

Procedure

1. Identify tactics and techniques.
2. Map indicators.
3. Export matrix.

Expected Results

- ATT&CK mapping matrix

Conclusion

ATT&CK improves detection strategy alignment.

Experiment 10: Threat Intelligence Report for SOC

Aim

To prepare a professional threat intelligence report.

Inputs

- Threat data

Tools Used

- Document editor
- Presentation tools

Dataset Required

- Real-world threat reports

Brief Theory

Threat reports inform SOC operations and leadership.

Procedure

1. Summarize the threat landscape.
2. Analyze impact.
3. Provide recommendations.

Expected Results

- SOC-ready threat report

Conclusion

Reports bridge technical and business perspectives.

Experiment 11: Configure Security Onion with Threat Feeds

Aim

To integrate threat feeds into Security Onion.

Inputs

- Threat feed URLs

Tools Used

- Security Onion
- Suricata

Dataset Required

- Threat intelligence feeds

Brief Theory

SIEM platforms enhance detection using external threat feeds.

Procedure

1. Configure feed ingestion.
2. Validate alerts.
3. Test detection.

Expected Results

- Generated threat alerts

Conclusion: Threat feeds increase detection accuracy.

Experiment 12: Threat Hunting with Zeek Using IOCs

Aim

To hunt threats using Zeek IDS.

Inputs

- PCAP files
- IOC list

Tools Used

- Zeek (Bro)
- Wireshark

Dataset Required

- Network traffic captures

Brief Theory

Threat hunting proactively searches for hidden threats.

Procedure

1. Load PCAP into Zeek.
2. Search for IOC matches.
3. Analyze traffic patterns.

Expected Results

- Identification of malicious traffic

Conclusion: Threat hunting improves detection beyond alerts.